

1 More on Primitive Natural Class Nominalism

The problem of relations The facts about the degrees of naturalness of all the classes of ordinary concrete particulars don't suffice to determine all the facts. EG: these facts don't capture the difference between the actual world with a possible world just like it except that everything is moved around.

A Realist would explain this difference as having to do with the question which *relations* are instantiated by which things. What can the PNCNist put in place of this?

One idea: we also need to know about the naturalness of *classes of classes* of particulars. EG: the set of all sets of two things which are exactly one metre apart might be natural (to a high degree).

A problem for this: how to analyse physically basic *non-symmetric* predicates?

Possible solution: posit *ordered pairs* (triples, etc.) in addition to classes.

Another possible solution: code pairs as sets of sets, e.g. using Kuratowski's method ($\langle a, b \rangle = \{a, \{a, b\}\}$).

Problem: arbitrariness.

Must PNCNists posit a property of naturalness? Armstrong suggests that they must, for reasons I don't understand. But this seems like a slip back into "loose talk".

2 Getting rid of classes?

PNCNism is not an option for the nominalist who wants to get rid of *classes* as well as properties and relations. But maybe we can regard all this talk of classes as just a manner of speaking?

We could take ‘natural’ to be, at bottom, a *plural* predicate. *The electrons* are natural. *The things that are electrons and within a mile of the surface of the Earth* are not.

3 Balance sheet: PNCNism

Big problem: how to define basic physical predicates, like ‘electron’?

Smaller problem: the “problem of relations”

4 “Ostrich” Nominalism

General idea: we look to empirical methods (physics?) to provide us with a list of primitive predicates; we use these predicates in defining notions of resemblance, naturalness, etc.

Perhaps we’ll come up with something like this:

- (1) For x to be a duplicate of y is for it to be the case that: x is an electron iff y is an electron, and x is a neutrino iff y is a neutrino, and ...

— Actually this is too simple, unless we’re Compositional Nihilists: we’ll want to allow for duplication between things that are too big to fall under basic physical predicates. An idea:

- (2) For x to be a duplicate of y is for there to be a bijection f between the parts of x and the parts of y , such that $f(x) = y$, and $f(z)$ is part of $f(w)$ iff z is part of w , and $f(z)$ is an electron iff z is an electron, and $f(z)$ is a neutrino iff z is a neutrino, and ..., and $f(z)$ is between $f(v)$ and $f(w)$ iff z is between v and w , and...

5 The problem of alien properties

Suppose for the moment that we're Realists. Presumably we'll think that, just as there could have been more or different *particulars* besides those there actually are, there could have been more or different *universals* besides those there actually are. For example, there are possible worlds where none of the universals that exist at the actual world exist, and instead there are some completely different universals, involved in some very different system of physical laws — e.g. laws of magic, or Aristotelian physics, or...

Even though a Nominalist can't accept the explanation in terms of universals, even a Nominalist might feel the force of the intuition that there are many interesting ways for the world to be in which there are no electrons, neutrinos...but rather things different in sort from any of these, and from each other.

- (3) Possibly, there are no electrons, and no neutrinos, and...and there are some things which are not exact duplicates of one another.

The Ostrich Nominalist must reject this claim.